

Course: B. Tech.

Branch : Information Technology

Semester : IV

Subject Code & Name: BTITC404, Design and Analysis of Algorithms

Max Marks: 60

Date: 27/12/2024

Duration: 3 Hrs.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and includes objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)	12
1 What is the worst case time complexity of merge sort?	1
a. $O(n \log n)$ b. $O(n^2)$ c. $O(\log n)$ d. $O(\log \log n)$	
2 Identify the sorting technique which compares adjacent elements in a list and switches whenever necessary?	1
a. Merge Sort b. Quick Sort c. Bubble Sort d. Selection Sort	
3 Which of the following algorithms is widely used for sorting large lists?	1
a. Merge Sort b. Bubble Sort c. Quick Sort d. Selection Sort	
4 What is the advantage of recursive approach than an iterative approach?	1
a. Consumes less memory b. Less code and easy to implement c. Consumes more memory d. More code has to be written	
5 The main time taking step in fractional knapsack problem is -----	1
a. Breaking items into fraction b. Adding items into knapsack c. Sorting d. looping through sorted items	
6 The multistage graph problem is to find the/a ____ path.	1
a. Maximum-cost b. Minimum-cost c. Shortest d. Longest	
7 Which of the following problems can not be solved using Greedy approach?	1
a. Huffman code b. Minimum spanning tree c. Job scheduling d. 0/1 Knapsack	
8 The files X1, X2, X3 are three files of length 30, 20, 10 records each. What is the optimal merge pattern value?	1
a. 110 b. 60 c. 90 d. 50	

- 9 Which of the following algorithms solve the Single Source Shortest Paths problem? 1
- a. Prim's algorithm b. Dijkstra's algorithm c. Bellman-Ford's algorithm d. Floyd-Warshall's algorithm
- 10 Breadth first search graph search algorithm uses ---- data structure for its implementation. 1
- a. Dequeue b. Tree c. Queue d. Stack
- 11 Topological sort can be applied to which of the following graphs? 1
- a. Undirected Cyclic Graphs b. Directed Cyclic Graphs c. Undirected Acyclic Graphs d. Directed Acyclic Graphs
- 12 What happens when the backtracking algorithm reaches a complete solution? 1
- a. It backtracks to the root b. It continues searching for other possible solutions c. It traverses from a different route d. Recursively traverses through the same route

Q. 2 Solve the following.

12

- A) What is an algorithm? (Understand, Apply/ CO1) 6
- Write an algorithm to find and return maximum of n given numbers.
- B) What is asymptotic notation? Explain Big "oh", Big Omega and Theta notations with suitable examples. <https://www.batuonline.com> (Understand/ CO2) 6

Q.3 Solve the following.

12

- A) What is a recurrence relation? How the complexity of algorithm is represented with recurrence relations? Give an example. (Apply/ CO4) 6
- B) What is divide and conquer strategy? (Understand, Create/ CO1) 6
- Write an algorithm for DandC Binary Search.

Q. 4 Solve Any Two of the following.

12

- A) Explain Strassen's method to compute the product of two $n \times n$ matrices. (Understand, Analyze/ CO4) 6
- Solve the recurrence relation for its computing time.
- B) State Knapsack problem. (Knowledge, Create/ CO1) 6
- Write greedy algorithm for solving it.
- C) What is the principle of optimality? (Knowledge, Create/ CO3) 6
- Write a dynamic programming formulation for a k-stage graph problem using the backward approach.

Q.5 Solve Any Two of the following.

12

- | | | | |
|----|---|--------------------------------|---|
| A) | Explain depth first search and traversal in the graph with a suitable example. | (Apply/
CO3) | 6 |
| B) | Write Kruskal's algorithm to generate a minimum cost spanning tree for connected undirected graph G. | (Create/
CO3) | 6 |
| C) | What is optimal storage on tape problem?
Write algorithm for tape storage problem if there are $M > 1$ tapes and n programs. | (Understand,
Apply/
CO1) | 6 |

Q.6 Solve Any Two of the following.

12

- | | | | |
|----|---|-------------------------------------|---|
| A) | Explain branch and bound technique.
Show how it differs from backtracking? | (Apply/
CO1) | 6 |
| B) | What is the concept of nondeterministic algorithm?
Write an algorithm for nondeterministic knapsack problem. | (Knowledge,
Create/
CO1, CO2) | 6 |
| C) | What are the constraints in Backtracking?
List the problems which can be solved using backtracking method. | (Understand/
CO1) | 6 |

***** End *****